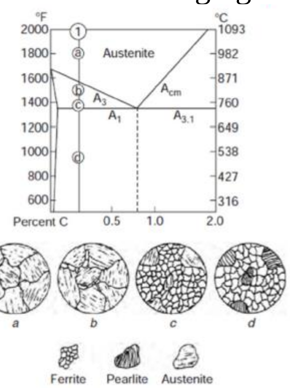
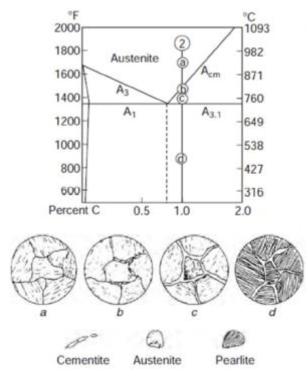
1. How many steps in a powder-metallurgy process? What are they?
2. What are the advantages and disadvantages of powder metallurgy technique?
3. Please give a brief definition to microstructure.
4. List the three allotropes of pure ion and their close-packing structure.
5. Please compare the properties of Fe-C alloy and pure Fe.
6. The eutectoid point is at what carbon content? At this point, what microstructure will form at room temperature? Also, please give a brief description to this microstructure.
7. Please order the following microstructures according their strength.
   Pearlite; martensite; bainite; tempered martensite; spheroidite
8. The main components of a stainless steel?

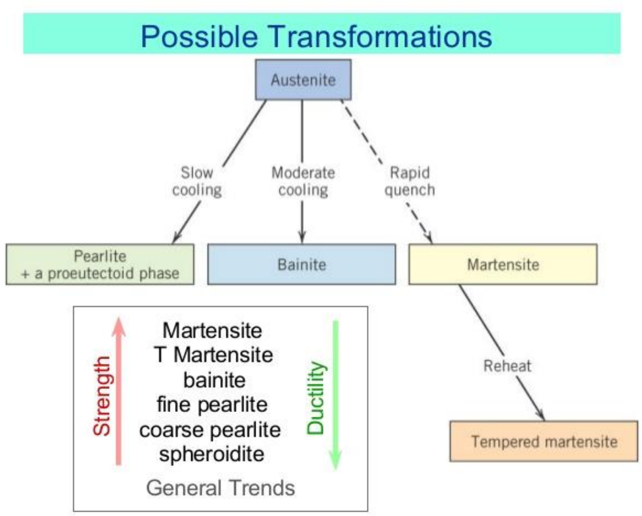
Chapter III

1. <u>Powder metallurgy</u>
   Five steps: powder production; blending; compaction; sintering; finishing operations.
2. <u>* Iron-Carbon Alloy</u>
   - Ferric (α -Fe); Austenite (γ -Fe); δ -Fe; Cementite (Fe_3C).
     In particular, understand that cementite is metastable phase.
   - Eutectoid steel (at 0.83 wt% a new microstructure, pearlite, formed): the phase diagrams of hypoeutectoid steel and hypereutectoid steel; Analyzing the microstructure changing during the cooling process.





3. <u>Hardening strategies for metals</u>
   - Grain size reduction, Solid-solution alloying, Work hardening, Precipitation hardening, Transformation hardening.
   - Martensite is the hardest and strongest of all possible Fe–C microstructures. Martensite forms when austenite is rapidly cooled (quenched) to room T. Since martensite is a metastable non-equilibrium phase, it does not appear in phase Fe-C phase diagram



4. <u>Stainless steel</u>
   An alloy of Fe, C and Cr.